Jianshu Zhao

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Research Interests

Large-scale genomics, metagenomics, microbiome analysis, long-read metagenomics, microbial ecology, metatranscriptomics, single-cell genomics, single-cell rRNA sequencing, genome search, biological network embedding, machine learning and deep learning in computational biology and high-performance computing.

Appointments

University of California, San Diego *2024–2026*
Postdoctoral Researcher

Education

Georgia Institute of Technology *2020–2024*
Ph.D. in Bioinformatics (minor in Computer Science & Engineering)

Tsinghua University *2016–2019*
M.S. in Environmental Science and Engineering

Publications

[*] *indicates first author or co-first author.*

Under Review and Preprints

[**P1**] **Zhao, J.**[*], McDonald, D., Patel, L., Wang, C., Gonzalez, A., and Knight, R. (2026). Homology-aware average nucleotide identity estimation at scale with TurboANI. *bioRxiv*

[**P2**] Din, M. O., Brennan, C.**, Zhao, J.**[*], Oles, R., Patel, L., Birmingham, A., Sevilir, D. M., Hansen, L., Wright, G. M., Ambre, M., Tomita, M., et al. (2026). Genome-resolved ecology and evolution of the microbiome enabled by quantitative long-read metagenomics. *bioRxiv*

[**P3**] **Zhao, J.**[*], Both, J. P., and Knight, R. (2025). Ultra-fast and Efficient Network Embedding for Gigascale Biological Datasets. *bioRxiv*

[**P4**] **Zhao, J.**[*], Zhao, X., Pierre-Both, J., and Konstantinidis, K. T. (2024). BinDash 2.0: New MinHash Scheme Allows Ultra-fast and Accurate Genome Search and Comparisons. *bioRxiv*

Published and Accepted

2026

[**J1**] **Zhao, J.**[*], McDonald, D., Sfiligoi, I., Lladser, M. E., Patel, L., Weng, Y., Khatib, L., Degregori, S., Gonzalez, A., Lozupone, C. A., and Knight, R. (2026). Megascale microbiome analysis with DartUniFrac. *Nature Biotechnology*

[**J2**] Khatib, L., Oles, R., Hernández, A. R., Myers, T.**, Zhao, J.**, Kumar, M., Laroya, A., Tolosa-Enguis, V., Matsunami, N., Suzuki, D., Chiba, N., et al. (2026). Five-country analysis of geographic and dietary drivers of gut microbiome composition and genomic variation. *Nature Communications*

[**J3**] Aton, M., McDonald, D., Cañardo Alastuey, J., Azom, R., Batra, P., Bezshapkin, V., Bolyen, E., Cagle, A., Caporaso, J. G., Debelius, J. W., Gorlick, K., Hamsanipally, N., Hunger, L., Keluskar, A., Liao, D., Lu, Y. Y., Navas-Molina, J. A., Pitman, A., Rideout, J. R., Sazonov, A., Sathappan, B., Schwarzberg Lipson, K., Sfiligoi, I., Tapo, C., Vázquez-Baeza, Y., Wu, Z., Xu, Z. Z., Ye, M. S.**, Zhao, J.**, Knight, R., Morton, J. T., and Zhu, Q. (2026). Scikit-bio: a fundamental Python library for biological omic data analysis. *Nature Methods* 23, 274–276

2025

[**J4**] **Zhao, J.***, Brandt, G., Gronniger, J. L., Wang, Z., Li, J., Hunt, D. E., Rodriguez-R, L. M., Hatt, J. K., and Konstantinidis, K. T. (2025). Quantifying the contribution of the rare biosphere to natural disturbances. *The ISME Journal* 19, wraf129

[**J5**] **Zhao, J.***, Pachiadaki, M., Conrad, R. E., Hatt, J. K., Bristow, L. A., Rodriguez-R, L. M., Rossello-Mora, R., Stewart, F. J., and Konstantinidis, K. T. (2025). Promiscuous and genome-wide recombination underlies the sequence-discrete species of the SAR11 lineage in the deep ocean. *The ISME Journal* 19, wraf072

[**J6**] Zhu, C., Wu, L., Ning, D., Tian, R., Gao, S., Zhang, B.**, Zhao, J.**, Zhang, Y., Xiao, N., Wang, Y., et al. (2025). Global diversity and distribution of antibiotic resistance genes in human wastewater treatment systems. *Nature Communications* 16, 4006

2024

[**J7**] **Zhao, J.***, Both, J. P., Rodriguez-R, L. M., and Konstantinidis, K. T. (2024). GSearch: ultra-fast and scalable genome search by combining K-mer hashing with hierarchical navigable small world graphs. *Nucleic Acids Research* 52, e74

[**J8**] **Zhao, J.***, Both, J. P., and Konstantinidis, K. T. (2024). Approximate nearest neighbor graph provides fast and efficient embedding with applications for large-scale biological data. *NAR Genomics and Bioinformatics* 6, lqae172

[**J9**] Qi, Q., Ning, S., Guo, X.**, Zhao, J.**, Tian, R., Gui, H., He, J.-S., Wang, H., Zhang, Z., Konstantinidis, K. T., Gao, Q., Wang, Y., Li, S., Zhao, W., Yang, Y., and Zhou, J. (2024). More sensitive microbial responses to the interactive effects of warming and altered precipitation in subsoil than topsoil of an alpine grassland ecosystem. *Global Change Biology* 30, e17487

2022

[**J10**] Menezes, O., Kocaman, K., Wong, S., Rios-Valenciana, E. E., Baker, E. J., Hatt, J. K.**, Zhao, J.**, Madeira, C. L., Krzmarzick, M. J., Spain, J. C., Sierra-Alvarez, R., Konstantinidis, K. T., and Field, J. A. (2022). Quinone Moieties Link the Microbial Respiration of Natural Organic Matter to the Chemical Reduction of Diverse Nitroaromatic Compounds. *Environmental Science & Technology* 56, 9387–9397

[**J11**] Qi, Q.**, Zhao, J.**, Tian, R., Zeng, Y., Xie, C., Gao, Q., Dai, T., Wang, H., He, J.-S., Konstantinidis, K. T., Yang, Y., Zhou, J., and Guo, X. (2022). Microbially enhanced methane uptake under warming enlarges ecosystem carbon sink in a Tibetan alpine grassland. *Global Change Biology* 28, 6906–6920

2021

[**J12**] Qi, Q., Haowei, Y., Zhang, Z., Van Nostrand, J. D., Wu, L., Guo, X., Feng, J., Wang, M., Yang, S., **Zhao, J.**, Gao, Q., Zhang, Q., Zhao, M., Xie, C., Ma, Z., He, J.-S., Chu, H., Huang, Y., Zhou, J., and Yang, Y. (2021). Microbial Functional Responses Explain Alpine Soil Carbon Fluxes under Future Climate Scenarios. *mBio* 12, e00761–20

2020

[**J13**] Wan, X., Gao, Q., **Zhao, J.**, Feng, J., Nostrand, J. D. van, Yang, Y., and Zhou, J. (2020). Biogeographic patterns of microbial association networks in paddy soil within Eastern China. *Soil Biology and Biochemistry* 142, 107696

2019

[**J14**] **Zhao, J.***, Gao, Q., Zhou, J., Wang, M., Liang, Y., Sun, B., Chu, H., and Yang, Y. (2019). The scale dependence of fungal community distribution in paddy soil driven by stochastic and deterministic processes. *Fungal Ecology* 42, 100856

[**J15**] Gao, Q., Yang, Y., Feng, J., Tian, R., Guo, X., Ning, D., Hale, L., Wang, M., Cheng, J., Wu, L., Zhao, M., **Zhao, J.**, et al. (2019). The spatial scale dependence of diazotrophic and bacterial community assembly in paddy soil. *Global Ecology and Biogeography* 28, 1093–1105

2015

[**J16**] Huang, Y., Sun, L., **Zhao, J.**, Huang, R., Li, R., and Shen, Q. (2015). Utilization of different waste proteins to create a novel PGPR-containing bio-organic fertilizer. *Scientific Reports* 5, 7766

[**J17**] Shen, Z., Zhong, S., **Zhao, J.**, Zhang, J., Li, R., Ruan, Y., and Shen, Q. (2015). Effects of ammonia fumigation on soil microflora and banana production in an orchard with serious Fusarium wilt disease. *Acta Ecologica Sinica* 35, 2946–2953

Presentations and Posters

- Poster, "Time-series metagenomics enables quantification of the rare biosphere to natural disturbances," ASM Microbe, Washington, DC, 2022.
- Poster, "High Performance parallel and concurrent programming for bioinformatics and computational biology," International Summer School on High Performance Computing, Atlanta, GA, 2023.
- Poster, "Promiscuous and unbiased recombination underlies the sequence-discrete species of the SAR11 lineage in the deep ocean," ASM Microbe, Atlanta, GA, 2024.

Skills

Compiled languages	Rust
Scripting and analysis	Python for GPU programming and modeling, R, statistics, modeling, visualization

Research Software

Key software and libraries: DartUniFrac; DartMinHash; TurboANI; GSearch; LASH; Annembed; GraphEmbed; Aeon; Rust-DiskANN; ExaLogLog; HyperMinHash; FastCover; SuperBits; Adas; BinDash-rs; Bindashtree; EMDUniFrac; fpcoa.

Contributed software and libraries: UltraLogLog; succparen; Scikit-bio; BinDash.